SDG BLOCKCHAIN ACCELERATOR

Market Penetration and Scaling Strategy

Fuel Switch - Renewable Energy Certification Platform
(Bosnia & Herzegovina)

This document outlines the market penetration strategy, branding approach, user acquisition plan, go-to-market strategy, and long-term scaling roadmap for the Fuel Switch pilot developed within the SDG Blockchain Accelerator program.

1. Project Overview

Fuel Switch is an Energy Attribute Certificate (EAC) management platform designed to enable transparent certification, management and trading of renewable energy attributes. The platform provides renewable energy producers and electricity traders with the tools required to issue, manage, transfer and retire Renewable Energy Certificates (RECs) and Guarantees of Origin (GOOs) through a secure digital platform supported by blockchain infrastructure.

The solution addresses a critical challenge currently emerging in Bosnia and Herzegovina's energy export market. Bosnia and Herzegovina is a net exporter of electricity and supplies power to several neighboring countries, including member states of the European Union. Beginning in January 2026, the European Union will implement the Carbon Border Adjustment Mechanism (CBAM), which introduces carbon taxation for imports whose production cannot demonstrate low-carbon or renewable origin.

At present, Bosnia and Herzegovina lacks a unified digital system capable of distinguishing renewable electricity from fossil-fuel-generated electricity within the national grid. Electricity generated from renewable sources and fossil fuels is transmitted through the same grid infrastructure, making it difficult to provide verifiable proof of renewable origin for exported electricity.

Without reliable certification mechanisms, electricity exporters risk having all exported electricity treated as non-renewable under CBAM regulations, which could result in significant financial penalties and loss of competitiveness in European markets.

Fuel Switch addresses this challenge by introducing a digital infrastructure capable of verifying renewable energy generation, issuing internationally recognized energy certificates, and enabling transparent tracking of renewable energy attributes across the electricity supply chain.

Through the SDG Blockchain Accelerator pilot, the platform demonstrates the complete lifecycle of renewable energy certification, including data verification, certificate issuance, certificate transfer, and retirement. The pilot includes real-world participants, including the renewable energy producer PV Romanovci and the electricity trader ELIFA-Soil, providing a realistic demonstration of the platform’s operational capabilities.

2. Market Penetration Strategy

2.1 Market Context and Opportunity

The renewable energy certification market in Bosnia and Herzegovina is expected to grow significantly in the coming years due to several key drivers. These include increasing investments in renewable energy infrastructure, growing international demand for verified green electricity, and new regulatory frameworks introduced by the European Union.

The implementation of the Carbon Border Adjustment Mechanism represents one of the most significant regulatory changes affecting electricity exports from the Western Balkans. Exporters must demonstrate the carbon intensity and renewable origin of electricity supplied to EU markets. Without appropriate certification infrastructure, electricity exporters risk facing carbon taxation or losing access to important export markets.

At the same time, Bosnia and Herzegovina possesses significant renewable energy potential, particularly in solar, hydro, and wind energy generation. As renewable energy production increases, the need for trusted certification infrastructure becomes increasingly important.

Currently, the national energy market lacks a comprehensive digital platform capable of managing renewable energy certificates in a transparent and verifiable manner. This creates a significant opportunity for digital platforms such as Fuel Switch to serve as enabling infrastructure for renewable energy certification.

By providing a trusted certification mechanism aligned with international standards, Fuel Switch supports both renewable energy producers and electricity traders in accessing global energy markets while complying with evolving regulatory requirements.

2.2 Target Market Segments

The platform serves several key user groups within the energy ecosystem.

The primary users of the platform are renewable energy producers operating solar, wind, hydroelectric, and other renewable generation facilities. These producers require certification mechanisms to verify the renewable origin of electricity and to monetize renewable attributes through renewable energy certificates.

Renewable energy certificates can provide an additional revenue stream for energy producers while also increasing the competitiveness of renewable electricity exports.

A secondary user group consists of electricity traders, utilities, and energy offtakers that purchase electricity from renewable energy producers. These organizations require verifiable certificates to export electricity to European Union markets, demonstrate compliance with CBAM regulations, and meet sustainability reporting requirements.

Industrial energy consumers also represent a potential future user group. Companies operating in carbon-intensive industries such as steel, aluminum, and manufacturing increasingly require verifiable renewable energy consumption to meet environmental reporting obligations and carbon reduction commitments.

In addition to commercial market participants, institutional stakeholders play an important role in enabling market adoption. These stakeholders include government ministries, regulatory authorities, and renewable energy registry operators responsible for energy market governance.

2.3 Market Entry Strategy

The market penetration strategy for Fuel Switch is structured around a phased approach designed to gradually build trust, demonstrate technical viability and scale adoption across the energy ecosystem.

Pilot Validation Phase

The first phase focuses on validating the concept through a real-world pilot implementation. The SDG Blockchain Accelerator pilot demonstrates the full lifecycle of renewable energy certification using actual market participants.

The pilot includes a renewable energy producer (PV Romanovci) and an electricity trader (ELIFA-Soil), representing a typical producer–offtaker relationship in the energy market. Through this pilot, the platform demonstrates the ability to verify generation data, issue renewable energy certificates, transfer certificates between market participants, and retire certificates as proof of renewable energy consumption.

This pilot phase plays a critical role in building trust with both regulators and market participants by demonstrating that the system can operate reliably in a real-world environment.

National Market Expansion

Following successful pilot validation, the next phase focuses on expanding adoption within the Federation of Bosnia and Herzegovina. This region currently lacks fully operational certification infrastructure, creating a strong opportunity for platforms capable of providing reliable certification mechanisms.

During this phase, Fuel Switch will onboard additional renewable energy producers and electricity traders who require certification services to access EU markets. Partnerships with

renewable energy operators and utilities will support broader adoption across the electricity market.

Regional Expansion

In the long term, Fuel Switch aims to expand beyond Bosnia and Herzegovina into other Western Balkan markets facing similar regulatory challenges. Countries such as Serbia, Montenegro, and North Macedonia also export electricity to EU markets and will be affected by CBAM requirements.

By establishing a successful implementation model in Bosnia and Herzegovina, Fuel Switch can replicate the certification infrastructure in these markets and contribute to regional renewable energy integration.

3. Branding and Positioning

Fuel Switch is positioned as trusted digital infrastructure for renewable energy certification, designed to support transparency, regulatory compliance, and international market access for electricity producers and traders. The platform addresses a critical gap in the renewable energy ecosystem by enabling reliable verification and management of renewable energy attributes, which are increasingly required for cross-border energy trade and sustainability reporting.

At its core, the Fuel Switch brand emphasizes trust, transparency, and accessibility. Renewable energy markets rely heavily on verifiable data and credible certification systems. By leveraging blockchain technology to create immutable records of renewable energy generation and certificate transactions, Fuel Switch provides an auditable system that enhances confidence among market participants, regulators, and international buyers. This transparency is particularly important in the context of emerging regulatory frameworks such as the European Union's Carbon Border Adjustment Mechanism (CBAM), which requires reliable proof of the carbon intensity and renewable origin of exported electricity.

The platform positions itself as an enabling infrastructure layer that simplifies complex certification processes and allows renewable energy producers and energy traders to participate more easily in international green energy markets. Rather than requiring producers to navigate fragmented certification systems and administrative processes, Fuel Switch provides a streamlined digital interface where renewable generation data can be verified, certificates issued, and transactions recorded in a transparent and secure manner.

Fuel Switch also differentiates itself by combining technological innovation with practical market integration. While blockchain technologies have been widely discussed in the energy sector, many solutions remain experimental or disconnected from real market needs. Fuel Switch focuses on practical deployment by integrating with existing energy market actors, regulatory frameworks, and renewable energy registries. This ensures that the platform complements existing energy market infrastructure rather than attempting to replace it.

Another important element of the platform's positioning is its role in supporting the modernization of energy markets in the Western Balkans region. Many countries in this region are undergoing rapid energy transition while simultaneously aligning their energy systems with European Union regulations. By providing transparent digital infrastructure for renewable energy certification, Fuel Switch contributes to improving market efficiency, enabling renewable investment, and supporting the region's integration into European energy markets.

Strategic partnerships further strengthen the credibility and positioning of the platform. Collaboration with organizations such as the United Nations Development Programme (UNDP) provides institutional legitimacy and demonstrates that the platform aligns with national development priorities and international sustainability goals. Through these partnerships, Fuel Switch positions itself not only as a technology provider but also as a trusted partner supporting energy transition, climate action, and sustainable economic development.

Ultimately, Fuel Switch aims to establish itself as a foundational infrastructure provider for renewable energy certification in emerging energy markets. By combining transparent technology, strong institutional partnerships, and practical market integration, the platform seeks to build long-term trust with both energy sector stakeholders and regulatory institutions.

4. User Acquisition and Go-To-Market Strategy

The user acquisition strategy focuses on building awareness and trust within the energy sector through institutional partnerships, industry engagement, and demonstration-based adoption.

Institutional partnerships play a central role in supporting adoption. Collaboration with government ministries, regulatory institutions, and international organizations ensures that the platform aligns with national energy policy and regulatory frameworks.

Key institutional partners include the UNDP Bosnia and Herzegovina country office, the Ministry of Foreign Trade and Economic Relations, the State Electricity Regulatory Commission, and renewable energy registry operators responsible for managing generation data.

Industry engagement will focus on renewable energy producers, electricity traders, utilities, and industrial energy consumers. These stakeholders represent the primary users of renewable energy certification infrastructure.

Fuel Switch will participate in renewable energy conferences, policy workshops, and stakeholder consultations to introduce the platform to potential users. These activities help educate market participants about the importance of renewable energy certification and demonstrate how digital platforms can simplify compliance processes.

The accelerator pilot itself serves as an important user acquisition tool. Demonstrating the full certification lifecycle using real market participants allows stakeholders to observe how the platform operates in practice. Pilot results will be documented and communicated through case studies and presentations shared with industry stakeholders.

5. Post-Accelerator Sustainability and Scaling Roadmap

5.1 Short-Term Implementation (0–6 Months)

Following completion of the accelerator program, the project will focus on transitioning the pilot system into a production-ready platform. This phase includes migrating the platform from the development environment to a production infrastructure, conducting smart contract security audits, and onboarding the first commercial users.

Initial commercial agreements will be established with pilot participants and early adopters within the renewable energy sector. These early transactions will demonstrate the commercial viability of the platform and provide reference use cases for broader market adoption.

5.2 Medium-Term Expansion (6–18 Months)

During the medium-term phase, the platform will expand across the Federation of Bosnia and Herzegovina. Additional renewable energy producers will be onboarded, and partnerships with electricity traders and utilities will be strengthened.

The project will also pursue regulatory recognition for renewable energy certificates issued through the platform. Collaboration with regulatory authorities will ensure that certificates align with national energy market regulations and international certification standards.

5.3 Long-Term Regional Growth (18+ Months)

In the long term, Fuel Switch aims to expand the certification model across the Western Balkans region. Countries such as Serbia, Montenegro, and North Macedonia face similar challenges related to renewable energy certification and CBAM compliance.

By establishing interoperable certification infrastructure across multiple countries, the platform can support regional renewable energy markets and contribute to the broader transition toward low-carbon energy systems.

6. Financial Sustainability Model

The long-term sustainability of the Fuel Switch platform will be supported through a multi-tier revenue model designed to scale alongside market adoption.

Revenue streams include onboarding fees for renewable energy facilities registering their generation assets within the platform, certificate issuance fees charged for the creation of renewable energy certificates, and transaction fees associated with certificate transfers and retirement.

This pricing model ensures accessibility for renewable energy producers of different sizes while enabling sustainable platform growth as the number of certified renewable energy transactions increases.

In addition to platform revenue, the project will pursue strategic partnerships and funding opportunities to support further development and regional expansion. Potential funding sources include climate technology investors, development finance institutions supporting renewable energy initiatives, and ecosystem-based funding programs within the blockchain industry.

Through this strategy, Fuel Switch aims to establish a scalable and sustainable infrastructure for renewable energy certification, supporting both national energy market modernization and regional integration into European renewable energy markets. The pilot developed through the SDG Blockchain Accelerator provides the foundation for broader adoption and long-term impact.